Geometry Note

Ran Xie

July 30, 2026

1 Definitions

Space of linear function $L(V, W)$ vector space of linear functions from V to W.

Dual Space $V^* = L(V, R)$. For each basis $\{e_i\}$ of V, there exists unique $\{e^i\}$ of V^* such that $e^i(e_j) = \delta^i_j$

Tensor Space $T^r_s = \underbrace{V \otimes V}_{\text{r times}} \otimes \underbrace{V^* \otimes V^*}_{\text{s times}}$ is space of multilinear functions on

$$\underbrace{V^* \times \ldots \times V^*}_{\text{r times}} \times \underbrace{V \times \ldots \times V}_{\text{s times}}$$

Tensor Product between A of (r, s) and B of (t, u), is

$$\begin{aligned} A \otimes B(\tau^1, \ldots, \tau^{r+t}, v_1, \ldots, v_{s+u}) \\ = A(\tau^1, \ldots, \tau^r, v_1, \ldots, v_s) \\ B(\tau^{r+1}, \ldots, \tau^{r+t}, v_{s+1}, \ldots, v_{s+u}) \end{aligned}$$

Vector Field X on coordinate neighborhood U of a manifold M, with coordinate x^i. For each point p, $X = X^i \partial_i$. $X[f] = X^i \partial_i f$

Change of Coordinates If Y has coordinate neighborhood V of y^i, then $Y^i = X^j \frac{\partial y^i}{\partial x^j}$

Map Differential(Pushforward) F_* is induced map $F_* : TM \to TN$ of C^∞ map $F : M \to N$. $F_*(v_p) = (F_* v)_{F(p)}$. With coordinate, $F_* = [\partial_j (y^i \circ F)]$, the Jacobian of F. Note that $y^i \circ F = F^i(x^1, \ldots, x^m)$

Tensor Bundle $T^r_s M$ of type (r, s) is the union of all tensor spaces $M^r_s(p)$ at each point $p \in M$.

Tangent Bundle $TM = T^1_0 M$,

Scalar Bundle $T^0_0 M = M \times \Re$,

Cotangent Bundle/ Differentials / Phase space $T^0_1 M$

Tensor Field T of type (r, s), $T(p) \in T^r_s M(p)$ for each p. $(1, 0)$ is vector field, $(0, 0)$ gives real-valued function. $(0, 1)$ gives differential.

Tensor Coordinate of T^r_s wrt coordinate x^i are d^{r+s} real-valued functions

$$T^{i_1 \ldots i_r}_{j_1 \ldots j_s} = T(dx^{i_1}, \ldots dx^{i_r}, \partial_{j_1}, \ldots, \partial_{j_s})$$

Tensor Product

Alternating Tensor: we can turn a k-tensor $T \in T^k(V^*)$ into an alternating tensor $\text{Alt}(T) \in \Lambda^k(V^*)$. by defining:

$$\text{Alt}(T) \equiv \frac{1}{k!} \sum_{\sigma \in S_k} (\text{sgn}\sigma)(T \circ \sigma)$$

where $S_k = \{\sigma(v_1, \ldots, v_k) = (v_{\sigma(1)}, \ldots, v_{\sigma(k)})\}$ is the group of all permutation of the arguments.

Wedge Product: After we construct the alternating tensors, we can define the wedge product between them.

$$T \wedge S = \binom{k+m}{k} \text{Alt}(T \otimes S)$$

where $T \in \Lambda^k(V^*)$, $S \in \Lambda^m(V^*)$ and $\text{Alt}(T \wedge S) \in \Lambda^{k+m}(V^*)$.

Pullback of a Covariant Tensor:

$$(f^*T)_p(v_1, \ldots, v_k) = T(f_* v_1, \ldots, f_* v_k) = T(df(v_1), \ldots, df(v_k))$$

Lie Derivative of a tensor field: Given a vector field X on M, the Lie derivative of a k-covariant tensor field T along X is

$$L_X T \equiv \frac{d}{dt}(\psi_t^* T)\Big|_{t=0}$$

, where $\psi_t = F(\cdot, t)$ with F the local flow of X at p.

REMARK: The intuition is simple, imagine T is a measurement device in a river placed around a neighborhood of point p and the water is flowing along a vector field X, we see that $\psi_t^* T(v_1, \ldots, v_k) = T(\psi_{t*} v_1, \ldots, \psi_{t*} v_k)$ which pushes forward the measurement device by t amount along the local flow around p. Then we take the limit as t goes to 0 like we do with usual function derivative to get the rate of change of the measurement device's reading along the flow.

Exterior Product

Differential forms p-form is C^∞ skew-symmetric covariant tensor field of degree p (type $(0, p)$). Local basis has $\binom{d}{p}$ p-forms $dx^{i_1} \cdots dx^{i_p}$ where $(i_1, \ldots, i_p)$ is increasing.

Vector Calculus via Differential Forms

Let (M, g) be a Riemannian 3-manifold. Let $\star$ be the Hodge star, d the exterior derivative, $\sharp : T^*M \to TM$ and $\flat : TM \to T^*M$ the musical isomorphisms, and $\omega_g = \star 1$ the Riemannian volume form.

Operation	**Nabla**	**Differential Form Expression**	**Remarks / Coordinate Formula (g_{ij})**
$\operatorname{grad} f$	∇f	$(df)^\sharp$	$(\operatorname{grad} f)^i = g^{ij}\partial_j f$
$\operatorname{div} X$	$\nabla \cdot X$	$\star d \star (X^\flat)$	$\operatorname{div} X = \frac{1}{\sqrt{\lvert g\rvert}}\partial_i(\sqrt{\lvert g\rvert}X^i)$
$\operatorname{curl} X$	$\nabla \times X$	$\big(\star d(X^\flat)\big)^\sharp$	$(\operatorname{curl} X)^i = \frac{1}{\sqrt{\lvert g\rvert}}\epsilon^{ijk}\partial_j(g_{k\ell}X^\ell)$
Δf	$\nabla^2 f$	$\star d \star df$	$\Delta f = \frac{1}{\sqrt{\lvert g\rvert}}\partial_i(\sqrt{\lvert g\rvert}g^{ij}\partial_j f)$

Musical Isomorphisms

For a vector field $X = X^i\partial_i$ and a 1-form $\alpha = \alpha_i dx^i$:

$$X^\flat = g_{ij}X^j dx^i,$$
$$\alpha^\sharp = g^{ij}\alpha_j \partial_i.$$

Hodge Star on $\mathbb{R}^3$ (Euclidean)

For standard dx, dy, dz:

$$\begin{aligned} \star 1 &= dx \wedge dy \wedge dz, \\ \star dx = dy \wedge dz, \quad \star dy &= dz \wedge dx, \quad \star dz = dx \wedge dy, \\ \star(dy \wedge dz) = dx, \quad \star(dz \wedge dx) &= dy, \quad \star(dx \wedge dy) = dz, \\ \star(dx \wedge dy \wedge dz) &= 1. \end{aligned}$$

Key Identities

- $\operatorname{curl}(\operatorname{grad} f) = 0 \iff d^2 = 0.$
- $\operatorname{div}(\operatorname{curl} X) = 0 \iff \star d(dX^\flat) = 0$
- $\operatorname{curl}(\operatorname{curl} X) = \operatorname{grad}(\operatorname{div} X) - \Delta X$ (Vector Laplacian). But I can skip this to keep it concise.

System	Coordinates / Length form / Volume form	Hodge $\star$ on 1-forms (Area forms)	Hodge $\star$ on 2-forms
Euclidean (Cartesian)	(x, y, z), $ds^2 = dx^2 + dy^2 + dz^2$ $\omega = dx \wedge dy \wedge dz$	$\star dx = dy \wedge dz$, $\star dy = dz \wedge dx$, $\star dz = dx \wedge dy$	$\star(dx \wedge dy) = dz$, $\star(dy \wedge dz) = dx$, $\star(dz \wedge dx) = dy$
Cylindrical	(ρ, ϕ, z), $ds^2 = d\rho^2 + \rho^2 d\phi^2 + dz^2$ $\omega = \rho\, d\rho \wedge d\phi \wedge dz$	$\star d\rho = \rho\, d\phi \wedge dz$, $\star d\phi = \frac{1}{\rho}\, dz \wedge d\rho$, $\star dz = \rho\, d\rho \wedge d\phi$	$\star(d\rho \wedge d\phi) = \frac{1}{\rho}\, dz$, $\star(d\phi \wedge dz) = \frac{1}{\rho}\, d\rho$, $\star(dz \wedge d\rho) = \rho\, d\phi$
Spherical	(r, θ, ϕ), $ds^2 =$ $dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta\, d\phi^2$ $\omega = r^2 \sin\theta\, dr \wedge d\theta \wedge d\phi$	$\star dr = r^2 \sin\theta\, d\theta \wedge d\phi$, $\star d\theta = \sin\theta\, d\phi \wedge dr$, $\star d\phi = \frac{1}{\sin\theta}\, dr \wedge d\theta$	$\star(dr \wedge d\theta) = \sin\theta\, d\phi$, $\star(d\theta \wedge d\phi) = \frac{1}{r^2 \sin\theta}\, dr$, $\star(d\phi \wedge dr) = \frac{1}{\sin\theta}\, d\theta$
Barycentric (2D)	(λ_1, λ_2), $\lambda_3 = 1 - \lambda_1 - \lambda_2$ $ds^2 =$ $g_{11} d\lambda_1^2 + 2 g_{12} d\lambda_1 d\lambda_2 + g_{22} d\lambda_2^2$, $g_{ij} = (\mathbf{v}_i - \mathbf{v}_3) \cdot (\mathbf{v}_j - \mathbf{v}_3)$ $\omega = \sqrt{G}\, d\lambda_1 \wedge d\lambda_2$, $G =$ $\det(g_{ij})$	$\star d\lambda_1 = \sqrt{G}\big(-g^{12} d\lambda_1 + g^{11} d\lambda_2\big)$, $\star d\lambda_2 = \sqrt{G}\big(-g^{22} d\lambda_1 + g^{12} d\lambda_2\big)$ where g^{ij} is the inverse of g_{ij}	$\star(d\lambda_1 \wedge d\lambda_2) = \frac{1}{\sqrt{G}}$

Table 1: Hodge star operator, length forms (ds^2), volume forms (ω), and area forms ($\star dx^i$) in common coordinate systems. Orientation is $(1, 2, 3)$ with $\epsilon_{123} = 1$; g^{ij} denotes the inverse metric. In barycentric coordinates, $\star d\lambda_i$ gives the area form of the face opposite vertex i, weighted by the metric.

1.0.1 Notations

Projection operator: $\langle\rangle_p : \bigwedge(V) \to \bigwedge_p(V)$,

$$\langle A\rangle_p = A_{[p]} \in \bigwedge_p(V)$$

is the p-vector component of the multivector A

- Grade Involution: $\sharp(A_{[p]}) = \widehat{A}_{[p]} = (-1)^p A_{[p]}$
- Reversion: $\widetilde{(\boldsymbol{v}_1 \wedge \cdots \wedge \boldsymbol{v}_p)} = \boldsymbol{v}_p \wedge \cdots \wedge \boldsymbol{v}_1$ implies $\widetilde{A}_{[p]} = (-1)^{p(p-1)/2} A_{[p]}$
- Conjugation: $\bar{A}_{[p]} = \widehat{\widetilde{A}}_{[p]} = \widetilde{\widehat{A}}_{[p]}$

1.1 Grassmann Algebra

Flat operator $\flat : \bigwedge_p(V) \to \bigwedge^p(V)$ given by

$$(\boldsymbol{v}_1 \wedge \cdots \wedge \boldsymbol{v}_p)_\flat = (\boldsymbol{v}_1)_\flat \wedge \cdots \wedge (\boldsymbol{v}_p)_\flat$$

Extension of the bilinear metric $G : \bigwedge_p(V) \times \bigwedge_p(V) \to \mathbb{R}$ given by

$$\begin{aligned} G(\boldsymbol{v}_1 \wedge \cdots \wedge \boldsymbol{v}_p, \boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_p) &= p!(\boldsymbol{v}_1 \wedge \cdots \wedge \boldsymbol{v}_p)_\flat(\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_p) \\ &= (\boldsymbol{v}_p \wedge \cdots \wedge \boldsymbol{v}_1)_\flat \rfloor (\boldsymbol{u}_1 \wedge \cdots \wedge \boldsymbol{u}_p) \\ &= \begin{Vmatrix} g(\boldsymbol{v}_1, \boldsymbol{u}_1) & \cdots & g(\boldsymbol{v}_1, \boldsymbol{u}_p) \\ \vdots & \ddots & \vdots \\ g(\boldsymbol{v}_p, \boldsymbol{u}_1) & \cdots & g(\boldsymbol{v}_p, \boldsymbol{u}_p) \end{Vmatrix} \end{aligned}$$

$G(A_{[p]}, B_{[q]}) = 0$ when $p \neq q$

Hodge Star Identities

Left Contraction

$$\begin{aligned} (\alpha \rfloor A_{[p]})(\alpha_1, \ldots, \alpha_{p-1}) &= pA_{[p]}(\alpha, \alpha_1, \ldots, \alpha_{p-1}) \\ &= p(\boldsymbol{v}_1 \wedge \ldots \wedge \boldsymbol{v}_p)(\alpha, \alpha_1, \ldots, \alpha_{p-1}) \qquad (\text{if } A_{[p]} = \boldsymbol{v}_1 \wedge \ldots \wedge \boldsymbol{v}_p) \\ &= p\frac{1}{p!}\sum_{\sigma \in S_p} \epsilon(\sigma)\alpha(\boldsymbol{v}_{\sigma(1)}) \cdots \alpha_{p-1}(\boldsymbol{v}_\sigma(p)) \\ &= \frac{1}{(p-1)!}\begin{vmatrix} \alpha(\boldsymbol{v_1}) & \ldots & \alpha(\boldsymbol{v_p}) \\ \alpha_1(\boldsymbol{v_1}) & \ldots & \alpha_1(\boldsymbol{v_p}) \\ \vdots & \ldots & \vdots \\ \alpha_{p-1}(\boldsymbol{v_1}) & \ldots & \alpha_{p-1}(\boldsymbol{v_p}) \end{vmatrix} \end{aligned}$$

So

$$(\alpha_1 \wedge \ldots \wedge \alpha_p)(\boldsymbol{v_1} \wedge \ldots \wedge \boldsymbol{v_p}) = \frac{1}{p!}(\alpha_1 \wedge \ldots \wedge \alpha_p) \rfloor (\boldsymbol{v_1} \wedge \ldots \wedge \boldsymbol{v_p}) = \begin{vmatrix} \alpha_1(\boldsymbol{v_1}) & \ldots & \alpha_1(\boldsymbol{v_p}) \\ \vdots & \ddots & \vdots \\ \alpha_p(\boldsymbol{v_1}) & \ldots & \alpha_p(\boldsymbol{v_p}) \end{vmatrix}$$

Properties

- $\alpha\rfloor(\boldsymbol{v}\wedge\boldsymbol{u}) = (\alpha\rfloor\boldsymbol{v})\boldsymbol{u} - \boldsymbol{v}(\alpha\rfloor\boldsymbol{u}) = \alpha(\boldsymbol{v})\boldsymbol{u} - \boldsymbol{v}\alpha(\boldsymbol{u})$
- $\alpha\rfloor(A\wedge B) = (\alpha\rfloor A)\wedge B + \hat{A}\wedge(\alpha\rfloor B)$ where $\hat{A}_{[p]} = \sharp A_{[p]} = (-1)^p A_{[p]}$. (Graded Leibniz rule)
- $(\alpha_1\wedge\ldots\wedge\alpha_p)\rfloor = \alpha_1\rfloor\alpha_2\rfloor\ldots\alpha_p\rfloor$

Orientation

For n linear independent vectors, $\boldsymbol{v_1}\wedge\ldots\wedge\boldsymbol{v_n} = J\boldsymbol{e_1}\wedge\ldots\wedge\boldsymbol{e_n}$. Orientation for V^* and V are

$$\Omega_{V^*} = J'\boldsymbol{e^1}\wedge\ldots\wedge\boldsymbol{e^n}, \qquad \Omega_V = J\boldsymbol{e_1}\wedge\ldots\wedge\boldsymbol{e_n}$$

They form two equivalent classes:

$$[\Omega] = \{A_{[n]}\in\bigwedge\nolimits_n(V) | A_{[n]} = J\Omega, \quad J>0\}$$
$$[-\Omega] = \{A_{[n]}\in\bigwedge\nolimits_n(V) | A_{[n]} = J\Omega, \quad J<0\}$$

Quasi-Hodge star

Given (V,Ω_V), define

$$\bar{\star}:\bigwedge\nolimits_p(V)\to\bigwedge\nolimits^{n-p}, \qquad \Psi^{[p]}\wedge\bar{\star}A_{[p]} = p!\Psi^{[p]}(A_{[p]})\Omega_{V^*}$$
$$\underline{\star}:\bigwedge\nolimits^p(V)\to\bigwedge\nolimits_{n-p}, \qquad A_{[p]}\wedge\underline{\star}\Psi^{[p]} = p!\Psi^{[p]}(A_{[p]})\Omega_V$$

Then we derive

$$\bar{\star}A_{[p]} = \tilde{A}_{[p]}\rfloor\Omega_{V^*}, \qquad \underline{\star}\Psi^{[p]} = \tilde{\Psi}^{[p]}\rfloor\Omega_V$$

When G is the Grassman extension of g and $\Omega_V = \frac{1}{\sqrt{|\det g|}}e_1\wedge\ldots\wedge e_n$(Pseudoscalar). Then we define hodge operator as composition of quasi hodge operator with the sharp/flat operator.

$$\star = \sharp\circ\bar{\star} = \underline{\star}\circ\flat$$

It holds under the condition that $G(\Omega_V,\Omega_V) = 1$ which will make $\sharp\circ\bar{\star} = \underline{\star}\circ\flat$.

Moreover

$$A\wedge\star B = G(A,B)\Omega_V, \qquad \star 1 = \Omega_V \qquad \star A = \tilde{A}_\flat\rfloor\Omega_V$$

For antiproduct(regressive product), we have

$$A_{[p]}\vee B_{[q]} = \star^{-1}(\star A_{[p]}\wedge\star B_{[q]})$$

The $\bigwedge^p(V)$ to $\bigwedge^{n-p}(V)$ version for forms is defined as map $\overset{*}{\star} = \flat\circ\star\circ\sharp$

Hodge Star Properties

Let (M^n, g) be an oriented Riemannian n-manifold with volume form $\omega = \star 1$. For $\alpha,\beta\in\Omega^k(M)$:

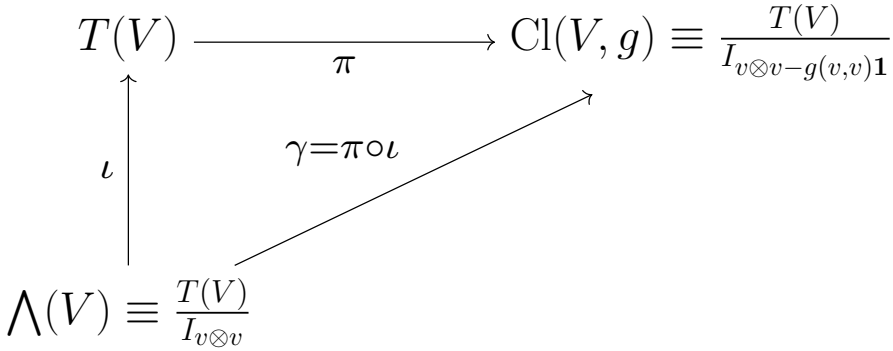
- **Distributive (linear):** $\star(\alpha+\beta) = \star\alpha + \star\beta$.
- **Double star:** $\star\star\alpha = (-1)^{k(n-k)}\alpha$.

- **Not commutative:** In general, $\star(\alpha \wedge \beta) \neq \star\alpha \wedge \star\beta$. Instead, the wedge product interacts via the metric:

$$\alpha \wedge \star\beta = \beta \wedge \star\alpha = \langle \alpha, \beta \rangle \, \omega,$$

  where $\langle \alpha, \beta \rangle = g^{i_1 j_1} \cdots g^{i_k j_k} \alpha_{i_1 \ldots i_k} \beta_{j_1 \ldots j_k}$.
- **Volume identities:** $\star 1 = \omega$, and $\star\omega = 1$.
- **Codifferential:** $\delta\alpha = (-1)^{n(k+1)+1} \star d \star \alpha$, satisfying $\delta = -d$ on Ω^0 (up to sign conventions).

1.2 Clifford Algebra



Here:

- $\pi : T(V) \to \mathrm{Cl}(V, g)$ is the Clifford projection
- $\iota : \bigwedge(V) \to T(V)$ is the inclusion map,
- $\gamma : \bigwedge(V) \to \mathrm{Cl}(V, g)$ is the bijective vector-space isomorphism **Clifford mapping**. It is natural to use the same notation used for exterior algebra for Clifford algebra. $\mathrm{Cl}(V, g) = \bigoplus_{p=0}^{n} \bigwedge_p(V)$ and projection operators is $\langle\rangle_p : \mathrm{CL}(V, g) \to \bigwedge_p(V)$ is equivalently defined.

Equivalently, γ is the Clifford algebra's "linear section" that identifies exterior wedge products with antisymmetrized Clifford products:

$$\gamma(v_1 \wedge \cdots \wedge v_k) = \frac{1}{k!} \sum_{\sigma \in S_k} \mathrm{sgn}(\sigma) \, v_{\sigma(1)} \cdots v_{\sigma(k)}.$$

1.2.1 Properties

- $\boldsymbol{v}\boldsymbol{u} = v \wedge u + g(\boldsymbol{v}, \boldsymbol{u})$
- $\boldsymbol{v}\boldsymbol{u} + \boldsymbol{u}\boldsymbol{v} = 2g(\boldsymbol{v}, \boldsymbol{u})$
- $\boldsymbol{v}A_{[p]} = \boldsymbol{v} \wedge A_{[p]} + \boldsymbol{v}_\flat \rfloor A_{[p]}$ (or $A_{[p]}\boldsymbol{v} = A_{[p]} \wedge \boldsymbol{v} + A_{[p]} \lfloor \boldsymbol{v}_\flat$)
- $\boldsymbol{v} \wedge A = \frac{1}{2}(\boldsymbol{v}A + \hat{A}\boldsymbol{v})$
- $\boldsymbol{v}_\flat \rfloor A = \frac{1}{2}(\boldsymbol{v}A - \hat{A}\boldsymbol{v})$

Cartan's Magic Formula (Lie derivative)

For a vector field X and a differential form $\alpha \in \Omega^k(M)$, the Lie derivative is:

$$\mathcal{L}_X \alpha = d(\iota_X \alpha) + \iota_X (d\alpha),$$

where ι_X denotes interior multiplication. For a vector field Y:

$$\mathcal{L}_X Y = [X, Y].$$

Cartan's Structure Equations

Let $\{e_i\}$ be a local orthonormal frame, $\{\theta^i\}$ its dual coframe, and $\omega^i{}_j$ the connection 1-forms defined by $\nabla_X e_j = \omega^i{}_j(X) e_i$ (Levi-Civita connection, torsion-free). The curvature 2-forms are $\Omega^i{}_j$.

Form versions

$$\text{First structure equation (torsion-free):} d\theta^i = -\omega^i{}_j \wedge \theta^j. \tag{1}$$

$$\text{Second structure equation (curvature):} \Omega^i{}_j = d\omega^i{}_j + \omega^i{}_k \wedge \omega^k{}_j. \tag{2}$$

The Riemannian curvature tensor satisfies $\Omega^i{}_j = \frac{1}{2} R^i{}_{jkl} \theta^k \wedge \theta^l$, i.e. $R(X, Y)Z = \iota_Z \Omega(X, Y)$ (index conventions may vary).

Vector field versions

For vector fields X, Y, Z:

$$\text{Torsion tensor (zero for Levi-Civita):} T(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y]. \tag{3}$$

$$\text{Curvature tensor (Riemann):} R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z. \tag{4}$$

For a torsion-free connection, Equation (3) simplifies to $\nabla_X Y - \nabla_Y X = [X, Y]$, which is exactly the dual of Equation (1).

Relation between form and vector versions

The connection 1-form relates to the covariant derivative via:

$$\omega^i{}_j(X) = \theta^i(\nabla_X e_j).$$

The curvature forms relate to the Riemann tensor by:

$$\Omega^i{}_j(X, Y) = \theta^i\big(R(X, Y)e_j\big).$$

Equivalently, $R(X, Y)Z = \iota_Z(\Omega^i{}_j \otimes e_i)$.

Maxwell's Equations: Nabla vs. Differential Forms

Definitions of the Differential Forms (Minkowski spacetime, metric $(-,+,+,+)$, SI units with $c = 1$)

Let $\mathbf{E} = E_x dx + E_y dy + E_z dz$ and $\mathbf{B} = B_x dx + B_y dy + B_z dz$ be the electric and magnetic field 1-forms. Define the **Faraday 2-form** F and the **current 3-form** J:

$$F = \mathbf{E} \wedge dt + \star_3 \mathbf{B} = E_x\, dx \wedge dt + E_y\, dy \wedge dt + E_z\, dz \wedge dt + B_x\, dy \wedge dz + B_y\, dz \wedge dx + B_z\, dx \wedge dy,$$

$$J = \rho\, \omega - \mathbf{J} \cdot d\mathbf{S} \wedge dt = \rho\, dx \wedge dy \wedge dz - (J_x\, dy \wedge dz + J_y\, dz \wedge dx + J_z\, dx \wedge dy) \wedge dt,$$

where $\omega = dx \wedge dy \wedge dz$ is the Euclidean volume form, $\star_3$ is the 3D Hodge star, and $d\mathbf{S} = (dy \wedge dz,\ dz \wedge dx,\ dx \wedge dy)$.

The Hodge star $\star$ in 4D gives the dual 2-form:

$$\star F = -\mathbf{E} \cdot d\mathbf{S} + \mathbf{B} \wedge dt = -E_x\, dy \wedge dz - E_y\, dz \wedge dx - E_z\, dx \wedge dy + B_x\, dt \wedge dx + B_y\, dt \wedge dy + B_z\, dt \wedge dz.$$

Side-by-Side Comparison

Law	Vector Calculus (Nabla) – SI	Differential Forms (SI with $c = 1$)
Gauss's law (electric)	$\nabla \cdot \mathbf{E} = \dfrac{\rho}{\epsilon_0}$	$d \star F = \mu_0 J$ (coefficient of ω)
Gauss's law (magnetic)	$\nabla \cdot \mathbf{B} = 0$	$dF = 0$ (coefficient of ω)
Faraday's law	$\nabla \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t}$	$dF = 0$ (coefficient of $d\mathbf{S} \wedge dt$)
Ampère-Maxwell law	$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \dfrac{\partial \mathbf{E}}{\partial t}$	$d \star F = \mu_0 J$ (coefficient of $d\mathbf{S} \wedge dt$)

Compact Unified Form

In differential forms, the four equations reduce to just two geometric statements:

$$\boxed{dF = 0} \qquad \text{(Bianchi identity / homogeneous equations)},$$

$$\boxed{d \star F = \mu_0 J} \qquad \text{(source equation / inhomogeneous equations)}.$$

Expanding $dF = 0$:

$$dF = (\nabla \cdot \mathbf{B})\, \omega \ + \ \left(\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} \right) \cdot d\mathbf{S} \wedge dt = 0,$$

which directly yields both $\nabla \cdot \mathbf{B} = 0$ and $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$.

Expanding $d \star F = \mu_0 J$:

$$d \star F = (\nabla \cdot \mathbf{E})\, \omega \ + \ \left(\nabla \times \mathbf{B} - \frac{\partial \mathbf{E}}{\partial t} \right) \cdot d\mathbf{S} \wedge dt = \mu_0 \rho\, \omega - \mu_0 \mathbf{J} \cdot d\mathbf{S} \wedge dt,$$

which yields both $\nabla \cdot \mathbf{E} = \rho / \epsilon_0$ (since $\mu_0 \epsilon_0 = 1$) and $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \partial_t \mathbf{E}$.

Reintroducing c and SI constants

If one wants to restore the speed of light c, define the Faraday 2-form with proper SI units:

$$F = \mathbf{E} \wedge (c\,dt) + \star_3 \mathbf{B} \quad \text{or} \quad F_{\text{SI}} = \frac{1}{c}\mathbf{E} \wedge dt + \star_3 \mathbf{B}.$$

With this normalisation, the equations become:

$$dF = 0, \qquad d \star F = \mu_0 J,$$

where J is defined exactly as above. Alternatively, in Gaussian units, the factor 4π shifts between F and J. The geometric structure $dF = 0,\ d \star F = J$ is universal; only the constants in the definition of J and F change.

Case Study 1: Surface of a sphere

The surface of sphere of radius 1 is a manifold

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 | x^2 + y^2 + z^2 = 1\}$$

We can define a chart (U, ψ) for S^2 where $U \subseteq M$ with spherical coordinate. Let

$$U = \{(\theta, \phi) \in [0, 2\pi] \times [0, \pi]\}$$

and

$$\psi(x, y, z) : \begin{cases} \theta = \arccos(z) \\ \phi = \text{sng}(y) \arccos \frac{x}{\sqrt{x^2+y^2}} \end{cases} , \psi^{-1}(\theta, \phi) : \begin{cases} x &= \sin\theta\cos\phi \\ y &= \sin\theta\sin\phi \\ z &= \cos\theta \end{cases}$$

Then $\psi(U) \subseteq \mathbb{R}^2$ is a homeomorphism from U to $\psi(U)$. ψ is called a **Locale coordinate map**. And the component functions (θ, ϕ) defined by $\psi(p) = (\theta(p), \phi(p))$ for $p \in S^2$ are called **local coordinates** on U.

One can think of this as giving a temporary identification between U and $\psi(U)$. When we work in this chart, we can think of U as an open subsets of the manifold and as an open subset of $\mathbb{R}^2$. Thus, we can represent a point $p \in U \subseteq S^2$ by its coordinate $(\theta, \phi) = \psi(p)$ and think of it as being the point p. We say (θ, ϕ) is the local coordinate for p or $p = (\theta, \phi)$ in local coordinates. *(See Lee's Smooth Manifold Local Coordinate Representations section)*

Given the same chart, the coordinate vectors $\partial_\theta, \partial_\phi$ form a basis for T_pS^2. If $v \in T_pS^2$, then

$$v = v^1 \frac{\partial}{\partial\theta}\bigg|_p + v^2 \frac{\partial}{\partial\phi}\bigg|_p = v^1\partial_\theta + v^2\partial_\phi = v^i\partial_i$$

The dual space to T_pS^2 is $T_p^*S^2$, if $w \in T_p^*S^2$,

$$w = w_1 d\theta + w_2 d\phi = w_i dx^i \text{(in generic coordinates)}$$

and $w(v) = w_i v^i$

S^2 is Riemannian with symmetric metric tensor defined as

$$\begin{aligned} g &= g_{ij} dx^i \otimes dx^j \\ &= g_{11} d\theta \otimes d\phi + g_{12} d\theta \otimes d\phi + g_{21} d\phi \otimes d\theta + g_{22} d\phi \otimes d\phi \\ &= g_{11}(d\theta)^2 + \frac{1}{2}(g_{12} + g_{21}) d\theta \otimes d\phi + \frac{1}{2}(g_{21} + g_{12}) d\phi \otimes d\theta + g_{22}(d\phi)^2 \qquad , (g_{12} = g_{22}) \\ &= g_{11}(d\theta)^2 + \frac{g_{12}}{2}(d\theta \otimes d\phi + d\phi \otimes d\theta) + \frac{g_{21}}{2}(d\phi \otimes d\theta + d\theta \otimes d\phi) + g_{22}(d\phi)^2 \\ &= g_{11}(d\theta)^2 + g_{12} d\theta d\phi + g_{21} d\phi d\theta + g_{22}(d\phi)^2 \\ &= g_{ij} dx^i dx^j \end{aligned}$$

We will now compute g. Since (θ, ϕ) are local coordinate of S^2, we can introduce a smooth immersion map $\iota = \psi^{-1} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$ into $\mathbb{R}^3$. Since $\mathbb{R}^3$ has Euclidean metric $\bar{g} = (dx)^2 + (dy)^2 + (dz)^2$, then g is the pullback of $\bar{g}$,

$$\begin{aligned} g &= \iota^* \bar{g} \\ &= (d(\sin\theta\cos\phi))^2 + (d(\sin\theta\sin\phi))^2 + (d(\cos\theta))^2 \\ &= (\cos\theta\cos\phi d\theta - \sin\theta\sin\phi d\phi)^2 + (\cos\theta\sin\phi d\theta + \sin\theta\cos\phi d\phi)^2 + (\sin\theta d\theta))^2 \\ &= (d\theta)^2 + \sin^2\theta (d\phi)^2 \end{aligned}$$

Case Study 2: Relativistic length contraction

Given a stationary frame, it has Minkowski flat metric of $(d\tau)^2 = (dt)^2 - (dx)^2$. A measure of length between location A and B along x-axis in a stationary frame are the distance between two simultaneous events $E_A = (t_0, x_a)^T$ and $E_B = (t_0, x_b)^T$. We want to calculate the distance with respect to a moving frame with constant velocity.

Let S be the stationary frame with axis (t, x) and S' with axis $(\bar{t}, \bar{x})$ be the moving frame in the x-direction with speed v. The line element $d\tau$ is invariant in different frames, therefore $(dt)^2 - (dx)^2 = (d\tau)^2 = (d\bar{t})^2 - (d\bar{x})^2$. So we have $(t)^2 - (x)^2 = (\bar{t})^2 - (\bar{x})^2$. The solution is given by

$$\begin{aligned} t &= \bar{t}\cosh\theta + \bar{x}\sinh\theta \\ x &= \bar{t}\sinh\theta + \bar{x}\cosh\theta \end{aligned}$$

The trajectory of the origin of S' along x axis is $x(t) = vt$ in frame S but $\bar{x}(\bar{t}) = 0$ in S' after the above transformation. So the above transformation is mapping $(\bar{t}, 0)$ to (t, vt). Substituting those in the solution, then we have $v = \tanh\theta \equiv \beta$. From that, we have $\cosh\theta = \frac{1}{\sqrt{1-v^2}} \equiv \gamma$, $\sinh\theta = \frac{v}{\sqrt{1-v^2}} = \gamma\beta$ and . We arrive at Lorentz transform from $S \to S'$:

$$\Lambda = \begin{pmatrix} \cosh\theta & \sinh\theta \\ \sinh\theta & \cosh\theta \end{pmatrix}^{-1} = \begin{pmatrix} \cosh\theta & -\sinh\theta \\ -\sinh\theta & \cosh\theta \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix}$$

Now consider the world line of A and B in S,

$$W_A(t) = \begin{pmatrix} t \\ x_a \end{pmatrix}, W_B(t) = \begin{pmatrix} t \\ x_b \end{pmatrix}$$

. After Lorentz transform,

$$\widehat{W}_A(t) = \begin{pmatrix} \gamma t - \gamma\beta x_a \\ -\gamma\beta t + \gamma x_a \end{pmatrix}, \widehat{W}_B(t) = \begin{pmatrix} \gamma t - \gamma\beta x_b \\ -\gamma\beta t + \gamma x_b \end{pmatrix}$$

As we can see when E_A and E_B that are simultaneous in S is not simultaneous in S' because their time component is not the same. To measure the distance in S', we will choose two events along the world line of A and B with the same time component in S' that is $\widehat{W}_A^{(0)}(t_a) = \widehat{W}_B^{(0)}(t_b)$. Therefore we have $\gamma t_a - \gamma\beta x_a = \gamma t_b - \gamma\beta x_b$. We can choose $t_a = \beta x_a$ and $t_b = \beta x_b$ which is their $\bar{x}$-intercept in S'.

Then the distance measured in S' is

$$\begin{aligned} \widehat{L} &= \widehat{W}_B^{(1)}(\beta x_b) - \widehat{W}_A^{(1)}(\beta x_a) \\ &= -\gamma\beta^2 x_b + \gamma x_b + \gamma\beta^2 x_a - \gamma x_a = \gamma(1-\beta^2)(x_b - x_a) \\ &= \frac{x_b - x_a}{\gamma} \\ &= \frac{L}{\gamma} \end{aligned}$$